

Research

Seasonal weather pattern prediction from enso indices using machine learning

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Abstract

Seasonal climate prediction in Bangladesh remains challenging due to the nonlinear nature of weather and climate interactions. This study investigates the correlation between nine El Niño–Southern Oscillation (ENSO) indices and seasonal temperature and rainfall patterns across Bangladesh, using monthly data from 29 meteorological stations (1977–2022). Six supervised machine-learning models, such as Random Forest (RF), XGBoost (XGB), Decision Tree (DT), Linear Regression (LR), K-Nearest Neighbors (KNN), and K-Fold Cross-Validation (KFCV) were evaluated using R^2 , MAE, and RMSE. XGB achieved the highest accuracy for temperature prediction ($R^2 = 0.8824$ for $T_{\max}$, 0.9706 for $T_{\min}$, and 0.9559 for T_{avg}), with RF and KFCV performing comparably. Rainfall prediction accuracy was lower, with RF achieving the highest R^2 (0.6273). Overall, the results confirm that multiple ENSO indices significantly influence Bangladesh's seasonal climate and that advanced ML models, particularly XGB and RF, offer strong potential for improved prediction.

Keywords Seasonal weather patterns prediction · ENSO · Supervised machine learning · Temperature · Rainfall

1 Introduction

Weather describes the short-term state of the atmosphere at a given place and time, whereas climate reflects the long-term (e.g., 30-year) average of those conditions [2]. Small variations in weather patterns can substantially influence temperature and precipitation enough to affect agriculture, hydropower, urban water management, and livelihoods [11]. Although short-range numerical models now provide reliable 7–10-day forecasts, their skill sharply declines beyond 2 weeks as initial-condition errors grow rapidly [28]. In contrast, seasonal forecasts based on slowly evolving large-scale climate drivers can offer communities and sectors in Bangladesh a critical window for preparedness [16]. Bangladesh is highly sensitive to climate anomalies, with nearly 80% of its annual rainfall occurring during the summer monsoon [1], often leading to severe flooding (e.g., 1987, 1988, 1998, 2007, 2024). Droughts and extreme cold events similarly disrupt socioeconomic conditions. Improving seasonal prediction capability is therefore essential for minimizing climate-related risks and enhancing regional resilience.

A major driver of climate variability is the El Niño–Southern Oscillation (ENSO), a coupled ocean–atmosphere phenomenon originating in the tropical Pacific with a recurrence period of 2–7 years [35]. ENSO's warm (El Niño) and cool

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(La Niña) phases modulate global rainfall, temperature, and circulation anomalies by altering sea-surface temperatures and atmospheric pressure gradients [12, 18]. These atmosphere–ocean anomalies are typically tracked using indices such as the Southern Oscillation Index (SOI), Oceanic Niño Index (ONI), and Niño 3.4 SST. Several other indices, including the Indian Ocean Dipole (IOD), Pacific Decadal Oscillation (PDO), Trans-Niño Index (TNI), Pacific North American (PNA) pattern, BEST, and upper-level zonal winds capture different aspects of regional and extra-tropical variability that often interact with ENSO. A concise summary of key indices, their regions, and phase characteristics is provided in the Supplementary Information.

Recent studies indicate that the performance of Niño 3.4 or ONI-based models has declined, partly due to changes in ENSO dynamics and the presence of the Spring Predictability Barrier [4]. South Asian studies show that rainfall is controlled by the combined influence of several climate indices, supporting the relevance of multi-index approaches [30, 31]. Although both dynamical and statistical models have been applied to ENSO-related forecasting [8, 23, 41], their skill varies substantially across lead times and regions. Machine-learning approaches have recently shown promise in capturing nonlinear interactions between teleconnection indices and regional climate variables [36], and Random Forest models, in particular, have demonstrated strong performance for rainfall prediction [33]. However, existing studies remain largely limited to single-index predictors or short-term forecasts.

Despite extensive work on ENSO-related variability, significant research gaps remain in current studies focusing on Bangladesh. Many studies rely on a single ENSO indicator, commonly Niño 3.4 SST or ONI, which captures only part of the coupled ocean–atmosphere system. This narrow approach often leads to inconsistent correlations, season-dependent sensitivities, and limited predictive skill. Furthermore, the combined influence of multiple ENSO-related indices, including IOD, PDO, TNI, BEST, PNA, and zonal wind anomalies, has not been systematically examined for Bangladesh, even though these drivers interact and may jointly modulate rainfall and temperature. Their interdependence raises additional questions about multicollinearity and the relative importance of each predictor.

To address these gaps, this study develops a data-driven framework that uses multiple ENSO-related indices to examine their combined influence on seasonal climate variability in Bangladesh. To comprehensively capture the role of large-scale ocean–atmosphere interactions, nine widely recognized ENSO-related and teleconnection indices are incorporated: ONI, NINO 3.4 SST, SOI, IOD, PDO, BEST, PNA, TNI, and 200 mb Zonal Winds. These indices were selected based on extensive climatological evidence demonstrating their influence on temperature anomalies, monsoon dynamics, and rainfall variability across South Asia and the broader Indo-Pacific region [3, 21, 32]. Each index represents a distinct physical mechanism—for instance, NINO 3.4 SST and ONI capture central Pacific sea-surface temperature anomalies, SOI reflects atmospheric pressure variability, IOD characterizes the Indian Ocean Dipole, while PDO, PNA, and TNI describe broader teleconnection modes that modulate ENSO impacts on regional climates [9, 13]. Because these indices often exhibit interdependence, potential multicollinearity was evaluated using pairwise correlations and variance inflation factors (VIF). To ensure robust predictive performance despite correlated predictors, ensemble machine learning models—specifically Random Forest and XGBoost—were chosen due to their demonstrated ability to handle multicollinearity and capture nonlinear interactions within high-dimensional climate data [7, 19]. This methodological design enables a more accurate estimation of how combined teleconnection signals influence rainfall and temperature variability across Bangladesh. Specifically, the study (i) quantifies the relative contributions of the nine teleconnection indices, (ii) evaluates the predictive skill of several ML algorithms (Random Forest, XGBoost, Decision Tree, Linear Regression, K-Nearest Neighbors) using MAE, RMSE, and R^2 , and (iii) examines how multicollinearity and feature importance affect model stability and interpretability. Through this integrated approach, the study provides an improved understanding of ENSO-related teleconnections and supports more reliable seasonal forecasting for Bangladesh.

2 Methodology

2.1 Dataset and study area

Four datasets of monthly maximum temperature, monthly minimum temperature, monthly average temperature and monthly average rainfall intensity have been processed using the data of 29 stations covering entire Bangladesh shown in Fig. 1. The stations are namely, Barisal, Bhola, Bogra, Chandpur, Chittagong, Comilla, Cox's Bazar, Dhaka, Dinajpur, Faridpur, Feni, Hatiya, Ishurdi, Jessore, Khepupara, Khulna, M.Court, Madaripur, Mymensingh, Patuakhali, Rajshahi, Rangamati, Rangpur, Sandwip, Satkhira, Sitakunda, Srimangal, Sylhet, Teknaf.

All the datasets also contain nine ENSO indices. They are ONI, NINO 3.4 SST, SOI, IOD, PDO, BEST, PNA, TNI, and 200mb Zonal Winds. Besides, each dataset also contains other parameters such as Year, Month, Season, Dominant Cycle, ENSO Event and its Intensity.

2.2 Data source

The ENSO indices were obtained from the NOAA Climate Prediction Center for the period 1977–2022. Temperature and rainfall data for the same period were collected from the Bangladesh Agricultural Research Council (BARC). Temperature is provided in degrees Celsius, and rainfall in millimeters.

2.3 Workflow

The research workflow, shown in Fig. 2, began with framing the problem and understanding the types of data required. Then, the required data were collected. Subsequently, data pre-processing was conducted to handle duplicate & missing values and remove outliers. After pre-processing, the exploratory data analysis (EDA) was conducted. In the EDA, Univariate, Bi-variate, and multi-variate analysis was carried out. EDA helps to understand the data, find the correlation among them and facilitate the model selection. Before feeding the data to train the model, data post-processing is required. Data post-processing is also known as ‘Feature Engineering’. Feature engineering was required for encoding categorical data and scaling the numerical data. Then, all the datasets were divided into two parts to train and validate the model. 70% of the total data were used to train the model and later, 30% of the total data were used to validate the model. Then, the models were assessed by using three performance metrics, namely Mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2) metrics. Finally, a comparison was demonstrated between actual scenario to predicted scenario, which highlighted the worthiness of the ML model.

2.4 Model selection

The Decision Tree (DT) regressor predicts the target variable by recursively partitioning the feature space into homogeneous subsets based on an optimal splitting criterion that minimizes within-node variance. For regression problems, the split at each node is selected by minimizing the sum of squared deviations of the target variable from its mean. The predictive structure of a decision tree can be expressed as a piecewise constant function over the input space. Model complexity is governed by parameters such as the maximum tree depth and the minimum number of samples required for node splitting and leaf formation, which directly control overfitting and stability. These parameters are particularly important when modeling climate variables, as meteorological datasets often contain noise and nonstationary [25].

$$VarSplit = \sum_{i=1}^N (y_i - \bar{y})^2 \quad (1)$$

where y_i is the observed value and $\bar{y}$ is the mean within a node.

Random Forest (RF) extends the decision tree framework by constructing an ensemble of trees trained on bootstrapped subsets of the data, with random feature selection at each split. The final prediction is obtained by averaging the outputs of all trees, expressed as:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (2)$$

where $h_t(x)$ represents the prediction of the t -th tree. Key model behavior is influenced by the number of trees, tree depth, and the number of predictors sampled at each split. These parameters collectively reduce variance and enhance generalization, making Random Forest particularly effective for capturing nonlinear and multivariate ENSO–climate relationships in seasonal forecasting applications [6].

Linear Regression (LR) models the dependent variable as a linear combination of predictor variables and serves as a baseline approach for assessing linear dependence between ENSO indices and climate variables. The model is defined as

$$y = \beta_0 + \sum_{i=1}^n \beta_i x_i + \varepsilon \quad (3)$$

where β_i are regression coefficients and ε denotes random error. Model performance depends on whether an intercept is included and whether regularization is applied to mitigate multicollinearity among predictors. Although simple, linear regression provides interpretability and acts as a reference against which more complex nonlinear models can be evaluated in climate modeling studies [37].

XGBoost is a boosting-based ensemble method that sequentially builds decision trees to minimize a regularized objective function combining prediction loss and model complexity. The optimization function is given by

$$\mathcal{L} = \sum_{i=1}^N l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (4)$$

where $\Omega(f_k)$ penalizes overly complex trees. Model performance is strongly influenced by hyperparameters such as learning rate, number of boosting iterations, maximum tree depth, and subsampling ratios. These parameters enable XGBoost to capture complex nonlinear interactions while controlling overfitting, which is essential for modeling climate variability driven by multiple interacting teleconnection indices [8, 22].

The K-Nearest Neighbors (KNN) algorithm is a non-parametric machine-learning technique used for both classification and regression tasks. Known for its simplicity and ease of implementation, KNN does not rely on assumptions about the data's underlying distribution. By identifying past seasons with comparable ENSO conditions to the target forecast season, this non-parametric analogue-based approach can make forecasts based on observable precedents rather than presumptive functional forms. For inter annual or seasonal weather prediction, KNN combines data mining to historical data and provides fairly accurate result.

$$\hat{y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i \quad (5)$$

Model behavior depends primarily on the choice of k and the distance metric used to identify nearest neighbors. In climate applications, KNN functions as an analogue-based forecasting approach, identifying seasons with similar ENSO conditions and inferring outcomes based on historical precedents rather than parametric assumptions [26].

K-Fold Cross-Validation evaluates model robustness by dividing the dataset into k subsets. The model is trained on $k-1$ folds and tested on the remaining fold. This process is repeated k times, each time with a different fold as the test set. The **CV score** is the **average performance metric across all folds**, which gives a more reliable estimate of model performance than using a single train-test split.

Mathematically:

$$CVScore = \frac{1}{k} \sum_{i=1}^k Metric_i \quad (6)$$

where k = number of folds, and $Metric_i$ = the evaluation metric (e.g., RMSE, MAE, R^2 , accuracy) calculated on the i -th fold. This method is particularly important in weather forecasting applications, where issues such as class imbalance and high variability can significantly influence predictive accuracy [14, 17].

Table 1 represents the value of different hyperparameters used in different models.

2.5 Model training and validation

This study utilized a data-driven methodology to examine the influence of 13 critical climate indices, including the Year, Month, ONI, NINO 3.4 SST, SOI, IOD, PDO, PNA, TNI, BEST, 200 mb Zonal Winds, ENSO Events, and their intensity. A data partitioning strategy was adopted, dividing the dataset into training (70%) and validation (30%) subsets, in line with standard practices in machine learning research [10, 38]. This splitting approach aimed to mitigate the risk of overfitting by ensuring the model's generalizability through performance assessment on unseen data [34].

The trained model was then subjected to rigorous evaluation on the independent validation set to measure its predictive accuracy and uncover potential biases. Such practices align with recommendations by leading machine learning

Table 1 represents the value of different hyperparameters used in different models

Model name	Hyperparameters with value
Decision Tree	max_depth=4, random_state=42
Random Forest	random_state=42
Linear Regression	No hyperparameter was used
XGBoost	max_depth=3, random_state=42, learning_rate=0.1, n_estimators=100
KNN	n_neighbors=5
K Fold Cross Validation	random_state=42, cv=5

Fig. 1 Geographical Distribution of Meteorological 29 Stations in Bangladesh

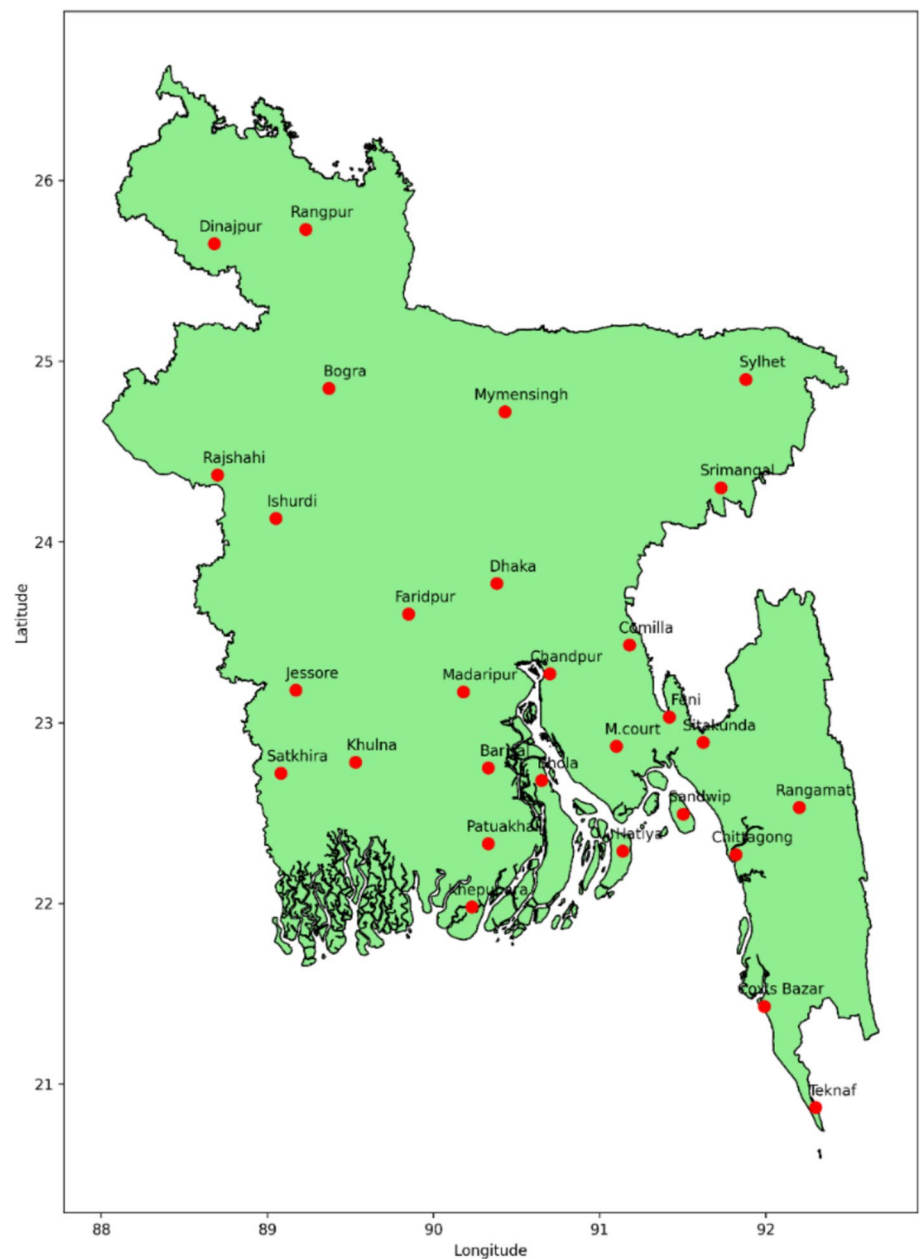
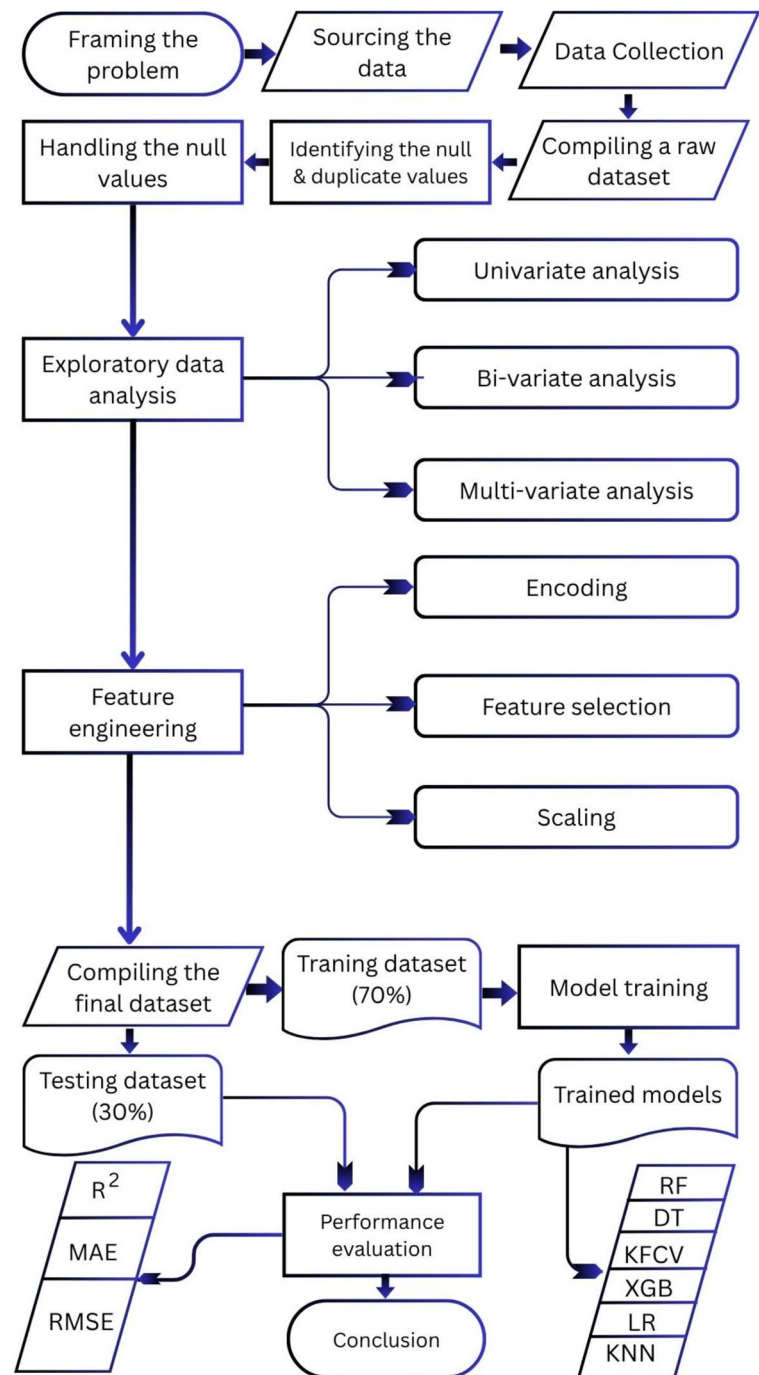


Fig. 2 Workflow Diagram

practitioners to ensure robust model validation and avoid spurious conclusions in predictive studies [40]. To evaluate the predictive performance of the machine learning models for seasonal temperature and rainfall forecasting, three widely used statistical metrics—Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2), were employed. MAE measures the average magnitude of prediction errors and provides a clear interpretation of overall model accuracy, which is particularly useful for assessing typical deviations in climate variables such as temperature and rainfall. RMSE emphasizes larger errors by squaring deviations, making it especially relevant for climate studies where extreme events (e.g., heavy rainfall or temperature extremes) have significant practical implications for flood risk management and climate resilience planning. R^2 quantifies the proportion of variance in the observed data explained by the model and is commonly used to assess how well multiple climate predictors collectively capture the variability of weather parameters. The combined use of MAE, RMSE, and R^2 ensures a comprehensive evaluation by

balancing overall accuracy, sensitivity to extremes, and explanatory power, which is essential in climate and hydrological modeling applications [27, 29].

By integrating MAE, RMSE, and R^2 , the study ensures a comprehensive evaluation of predictive performance for both temperature and rainfall. MAE evaluates overall accuracy, RMSE emphasizes sensitivity to extremes, and R^2 assesses how well multiple input indices explain the variability of each target variable. This approach is critical in climate modeling and hydrological applications, where accurate predictions of temperature and rainfall are essential for water resource management, agriculture, and disaster preparedness [20].

3 Results and analysis

3.1 Data pre-processing

During the data preprocessing phase, we identified a few instances of null values within the dataset. However, no duplicate entries were detected. Given the limited number of null values, we opted to remove the corresponding rows entirely to ensure data integrity. This was achieved using the `dropna` function provided by the Pandas library, which allowed us to eliminate rows containing null values efficiently. This approach preserved the overall quality and consistency of the dataset while minimizing the impact of missing data on subsequent analyses. After completing the data cleaning process, the temperature datasets were reduced to dimensions of 539 rows and 47 columns, while the rainfall dataset was reduced to dimensions of 527 rows and 47 columns.

3.2 Exploratory data analysis (EDA)

To gain a comprehensive understanding of the dataset, exploratory data analysis (EDA) was conducted. This study involved univariate, bivariate, and multivariate analyses to examine the data from different perspectives and identify underlying patterns.

3.2.1 Univariate analysis

For categorical variables, such as ENSO Event, and Season, monthly count plots were generated to analyze their frequency distributions. For numerical variables, Kernel Density Estimate (KDE) plots were utilized to examine the data distribution.

Figure 3 highlights La Nina events occur more frequently than El Nino and Neutral events, showing variability in the ENSO Events.

On the other hand, Fig. 4 shows the Neutral events have the highest frequency, followed by Weak El Nino. Besides, Very Strong El Nino have much lower counts, indicating they are less frequent.

Fig. 3 Monthly Count Plot of ENSO Event

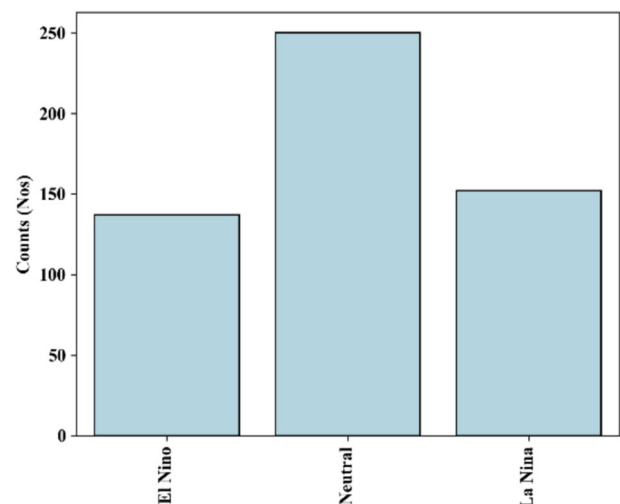


Figure 5 represents the Neutral–Neutral phase condition is dominant with a higher frequency compared to remaining conditions. This suggests a higher occurrence of Neutral–Neutral conditions in the dataset and highlights the count of occurrences for different seasons. The El Nino-Neutral, Neutral-El Nino, Neutral-La Nina, La Nina-Neutral has the lowest count.

Figure 6 shows that ONI values are concentrated around zero, indicating the dominance of neutral ENSO conditions with fewer extreme El Niño or La Niña events.

Fig. 4 Monthly Count Plot of ENSO Event with Intensity

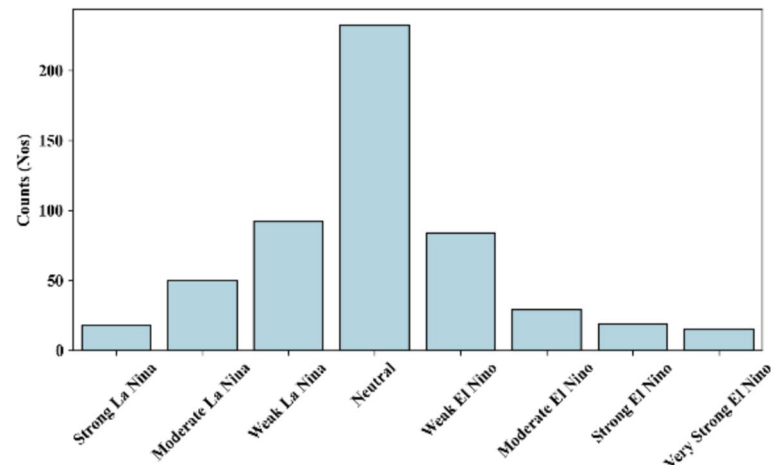


Fig. 5 Monthly Count Plot of Phase Conditions

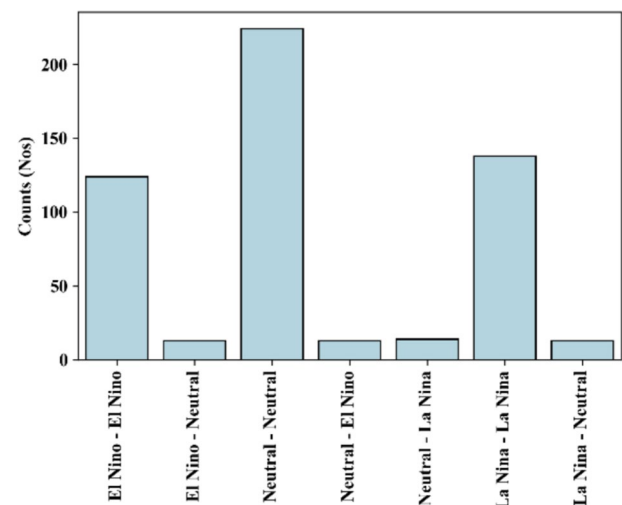


Fig. 6 KDE Plot of ONI

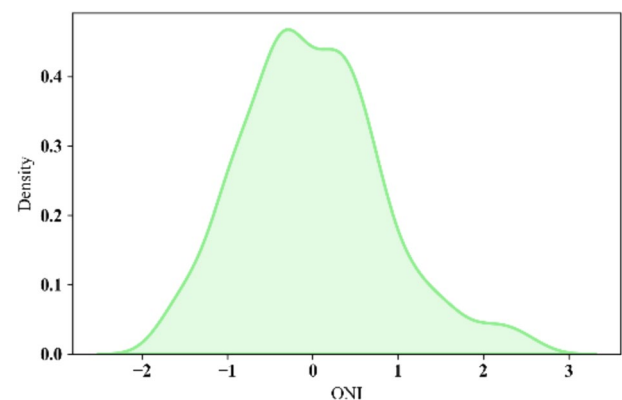


Figure 7 illustrates that NINO 3.4 SST anomalies are mostly centered near zero, suggesting balanced warm and cool phase occurrences over the study period.

Figure 8 indicates a near-symmetric distribution of SOI values, reflecting comparable frequencies of positive and negative atmospheric pressure anomalies.

Figure 9 shows that IOD values cluster around zero, implying that neutral Indian Ocean Dipole conditions are most common.

Figure 10 highlights that PDO values are largely concentrated near zero, consistent with its low-frequency, decadal-scale variability.

Figure 11 demonstrates that BEST index values are centered around zero, indicating limited dominance of extreme SST anomalies.

Figure 12 shows a symmetric distribution of PNA values, suggesting balanced positive and negative teleconnection phases.

Figure 13 indicates that TNI values are mostly near zero, reflecting weak-to-moderate trans-Pacific SST variability.

Fig. 7 KDE Plot of NINO 3.4 SST

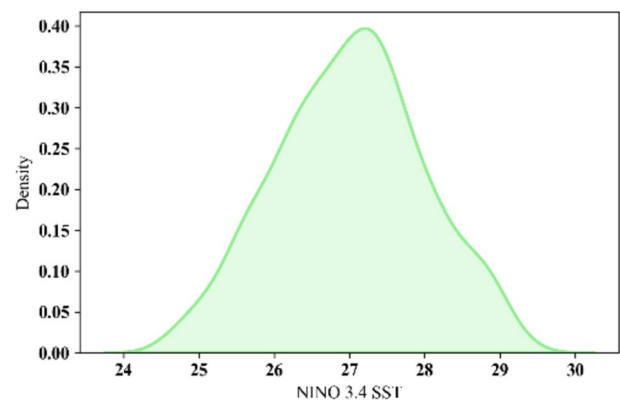


Fig. 8 KDE Plot of SOI

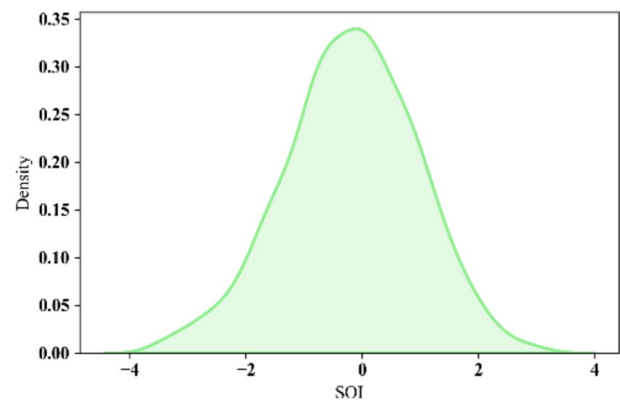


Fig. 9 KDE Plot of IOD

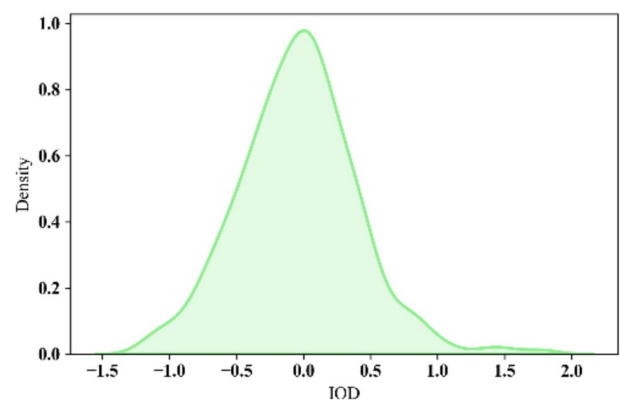


Fig. 10 KDE Plot of PDO

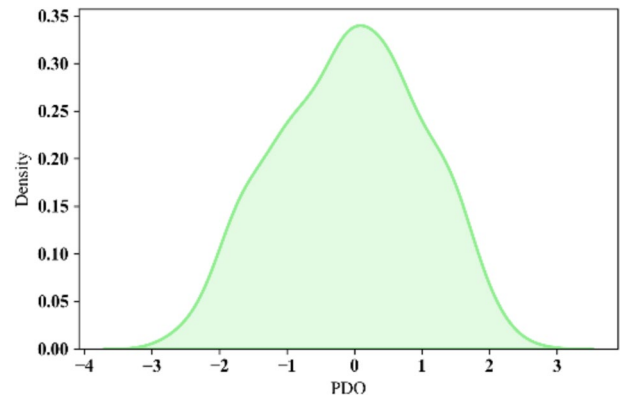


Fig. 11 KDE Plot of BEST

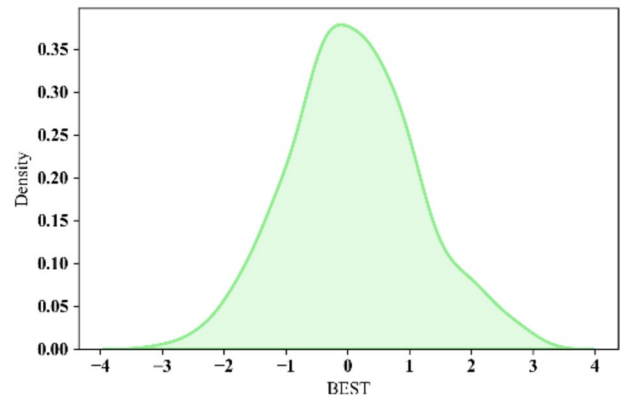


Fig. 12 KDE Plot of PNA

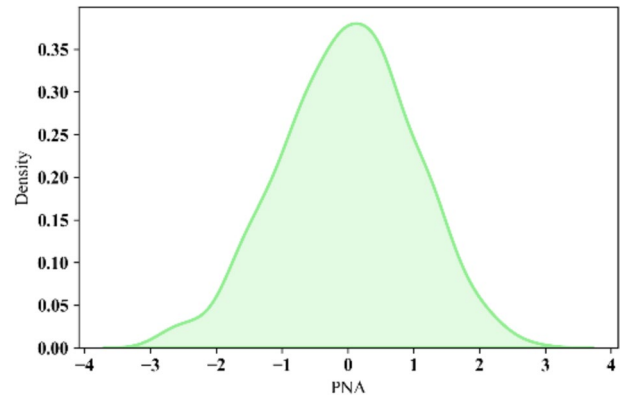


Fig. 13 KDE Plot of TNI

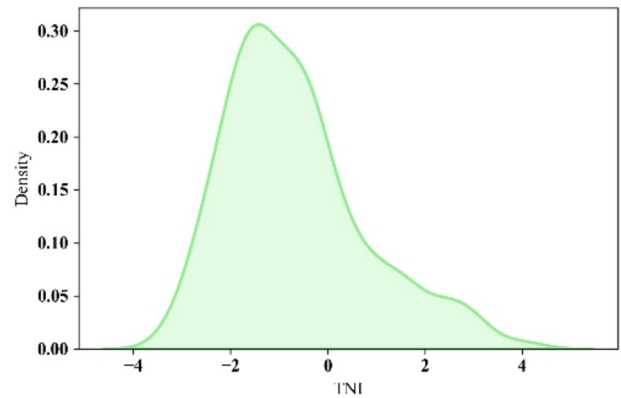


Figure 14 shows that zonal wind anomalies are evenly distributed around zero, indicating stable upper-level circulation conditions.

3.2.2 Bi-variate analysis

Figures 15, 16, 17 and 18 illustrates the distribution of ENSO events (El Nino, Neutral, and La Nina) across four seasons: Winter, Pre monsoon, Monsoon, and Post monsoon. In all seasons except Post monsoon, Neutral events exhibit the

Fig. 14 KDE Plot of 200mb Zonal Winds

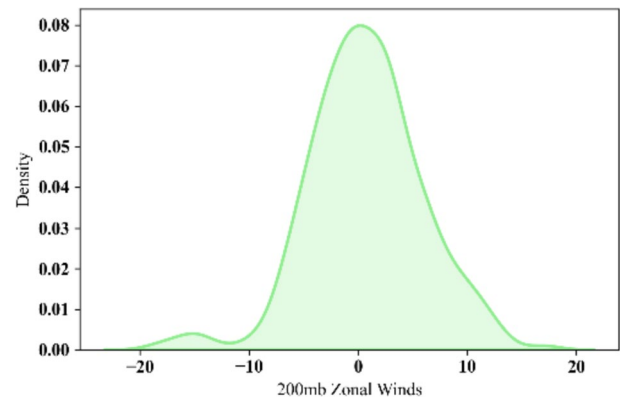


Fig. 15 Bivariate Analysis between Winter Season and ENSO Event

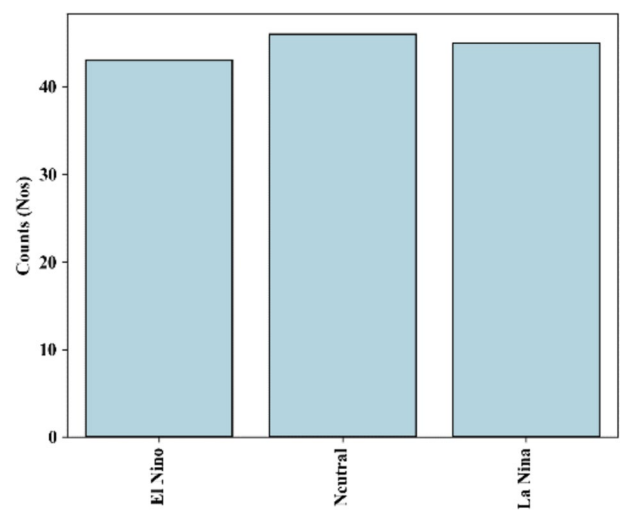


Fig. 16 Bivariate Analysis between Pre Monsoon Season and Enso Event

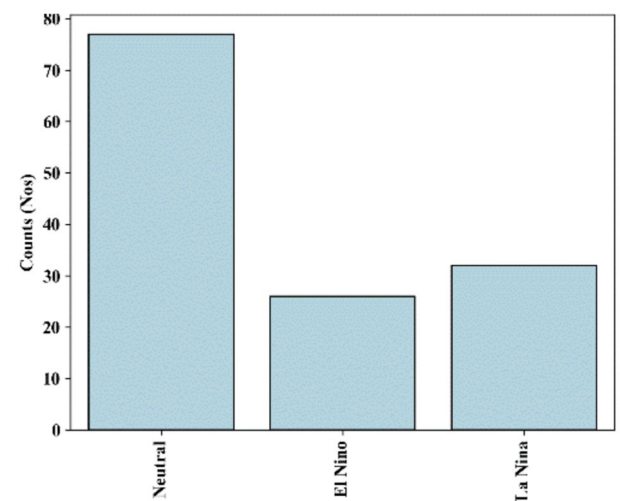


Fig. 17 Bivariate Analysis between Monsoon Season and Enso Event

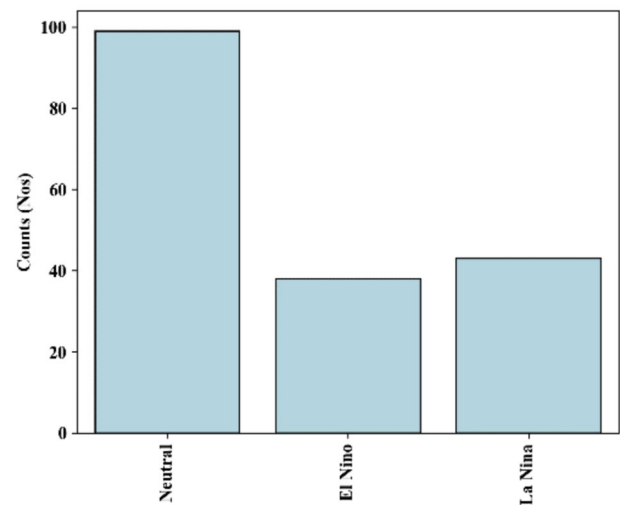
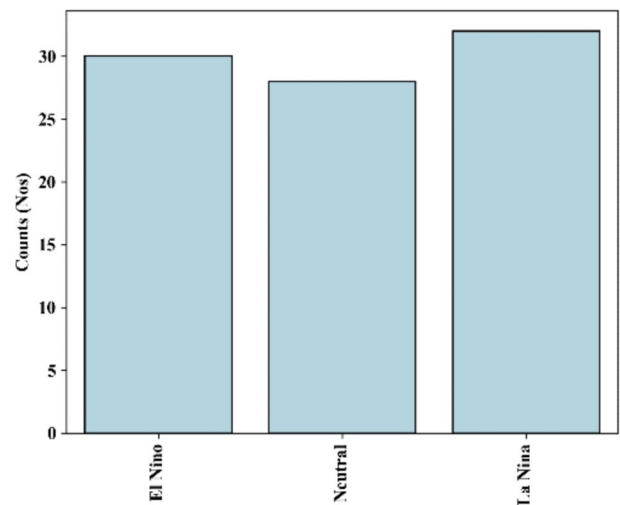


Fig. 18 Bivariate Analysis between Post Monsoon Season and Enso Event



highest frequency, while El Nino events are the least common. But, La Nina events have the highest frequency in the Post monsoon.

Figure 15 illustrates the distribution of ENSO events during the winter season. Neutral conditions dominate winter months, indicating relatively stable large-scale climate conditions. El Niño occurrences are least frequent, while La Niña events appear moderately, suggesting limited but notable cold-season ENSO influence.

Figure 16 shows the frequency of ENSO events during the pre-monsoon season. Neutral events remain the most common, reflecting transitional atmospheric conditions. La Niña events occur more frequently than El Niño, indicating a tendency toward enhanced variability prior to monsoon onset.

Figure 17 presents the distribution of ENSO events during the monsoon season. Neutral phases are again dominant, while La Niña events occur more often than El Niño events. This pattern highlights the stronger association of La Niña with monsoon-season variability.

Figure 18 depicts ENSO event frequencies in the post-monsoon season. Unlike other seasons, La Niña events are most frequent, suggesting a stronger ENSO signal during monsoon withdrawal. Neutral conditions decrease, indicating increased climate variability in this period.

Figure 19 illustrates ENSO phase transitions during winter. Most observations remain within the same phase, indicating phase persistence. No direct transitions between El Niño and La Niña are observed, reflecting gradual seasonal evolution of ENSO conditions.

Fig. 19 Bivariate Analysis between Winter Season and Phase Condition

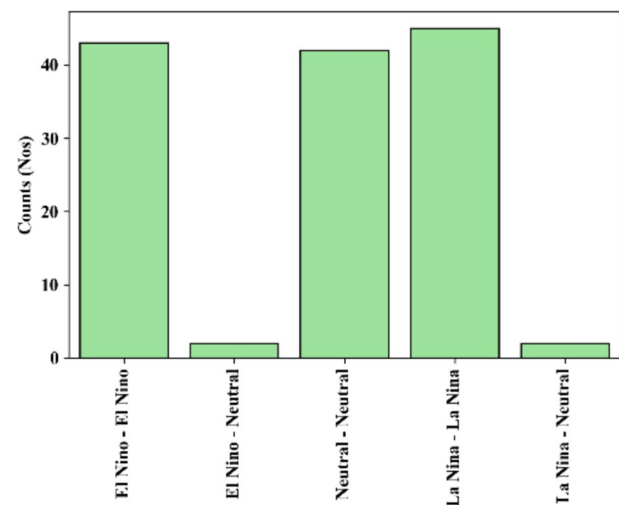


Fig. 20 Bivariate Analysis between Pre Monsoon Season and Phase Condition

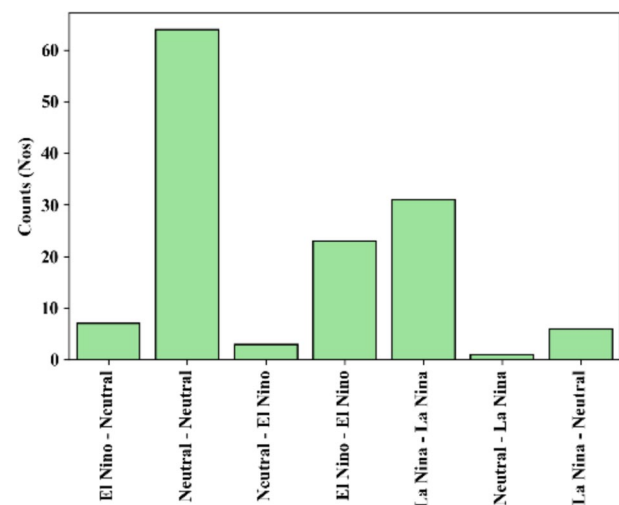


Figure 20 shows ENSO phase conditions during the pre-monsoon season. Stable phase conditions dominate, with limited transitions. Changes, when present, mainly occur between Neutral and ENSO-active states, highlighting gradual pre-monsoon adjustments.

Figure 21 presents ENSO phase behavior during the monsoon season. The results indicate strong phase stability, with minimal transitions. This persistence suggests that established ENSO conditions tend to maintain their influence throughout the monsoon period.

Figure 22 illustrates phase conditions during the post-monsoon season. While phase stability remains dominant, slightly more transitions are observed compared to other seasons. However, direct El Niño–La Niña shifts are still absent, confirming gradual ENSO evolution.

3.2.3 Multi-variate analysis

In this study, multivariate analysis is applied to explore the relationships between ENSO indices and weather parameters across 29 stations. To gain more specific insights, the datasets for $T_{\max}$, $T_{\min}$, T_{avg} , and Rainfall were divided into eight ENSO events: Weak El Nino, Moderate El Nino, Strong El Nino, Very Strong El Nino, Neutral, Weak La Nina, Moderate La Nina, and Strong La Nina, facilitating a detailed examination of event-specific correlations. This approach enhances the precision of the analysis, uncovering complicated associations that deepen the understanding of how ENSO indices influence regional climate variability. The heatmap uses a seven-color scale to represent the strength and direction of correlations: dark green denotes strong negative correlations (-1 to -0.7), medium green indicates

Fig. 21 Bivariate Analysis between Monsoon Season and Phase Condition

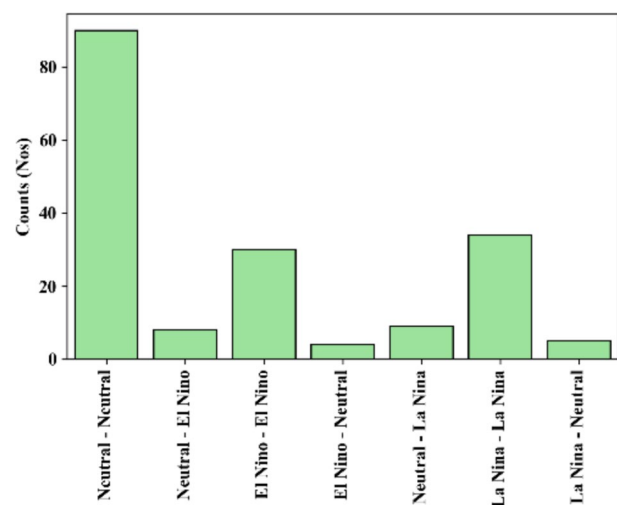


Fig. 22 Bivariate Analysis between Post Monsoon Season and Phase Condition

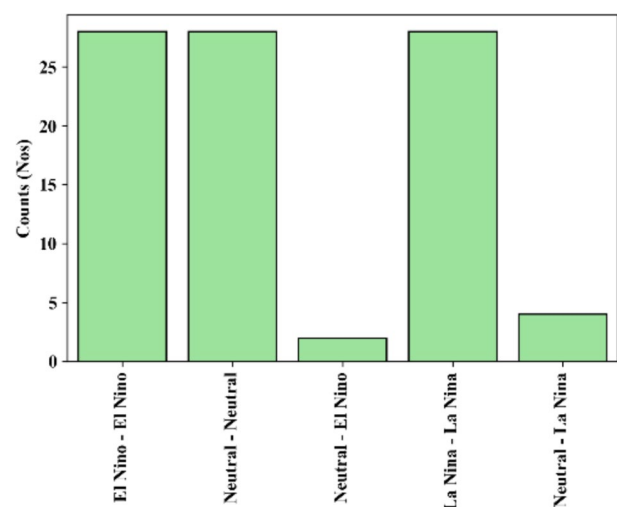
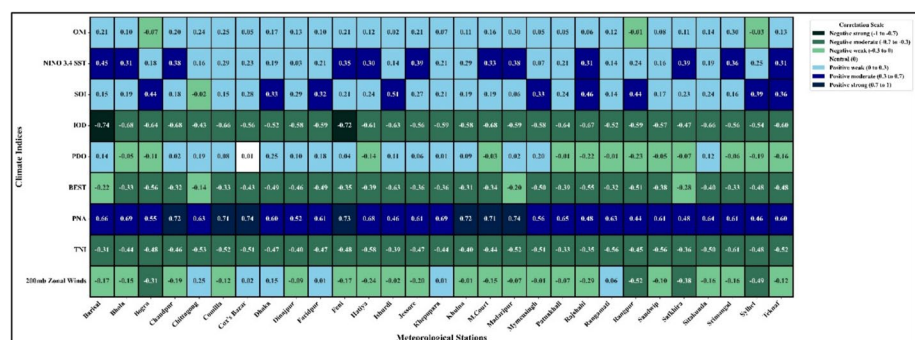


Fig. 23 Correlation of ENSO Indices with Rainfall Across 29 Stations in Bangladesh during Strong La Nina Events



moderate negative correlations (-0.7 to -0.3), light green represents weak negative correlations (-0.3 to 0), white denotes neutral relationships (0), light blue corresponds to weak positive correlations (0 to 0.3), medium blue reflects moderate positive correlations (0.3 to 0.7), and deep blue captures strong positive correlations (0.7 to 1).

Figure 23 shows the correlation between various climate indices and rainfall across different regions during strong La Nina events. The x-axis lists the meteorological stations across Bangladesh, representing regional variations in rainfall responses, while the y-axis presents the major climate indices, including ONI, NINO 3.4 SST, SOI, IOD, PDO, BEST, PNA, TNI, and 200 mb Zonal Winds. Notably, IOD and TNI exhibit strong negative correlations in most regions, indicating reduced

Fig. 24 Correlation of ENSO Indices with T_{min} Across 29 Stations in Bangladesh during Strong La Nina Events

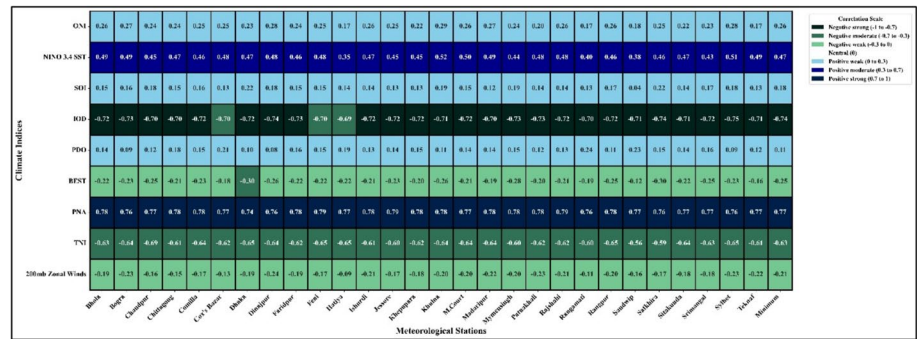
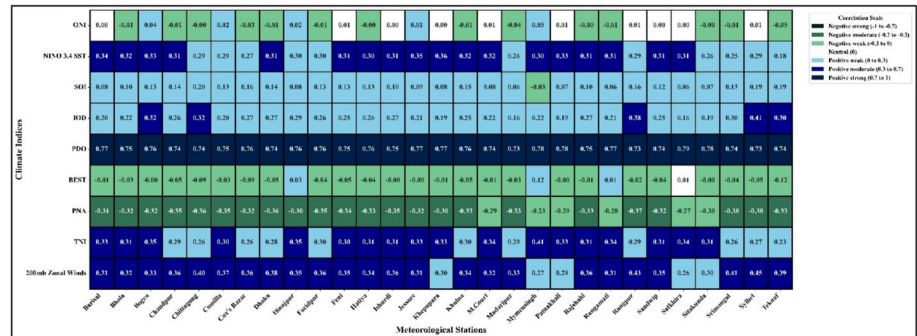


Fig. 25 Correlation of ENSO Indices with T_{avg} Across 29 Stations in Bangladesh during Strong El Nino Events



rainfall influence. PNA shows a consistently strong positive correlation, especially in southeastern areas. NINO 3.4 SST also shows moderate positive correlations. These patterns suggest that during strong La Nina, regional rainfall is more strongly influenced by IOD, PNA, and TNI.

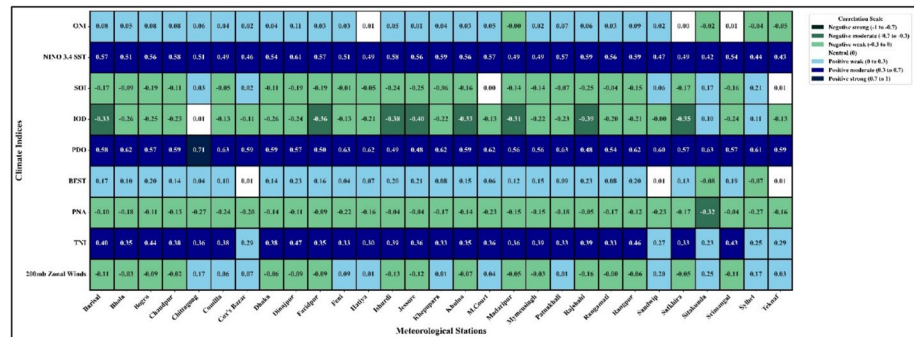
Figure 24 shows strong correlations between climate indices and minimum temperature during strong La Nina events. The x-axis displays meteorological stations across Bangladesh, capturing spatial variations in minimum temperature patterns, while the y-axis includes the key climate indices—such as ONI, NINO 3.4 SST, SOI, IOD, PDO, BEST, PNA, TNI, and 200 mb Zonal Winds—that influence T_{min} variability. PNA exhibits a consistently strong positive correlation across all regions, while IOD and TNI show strong negative correlations, indicating significant influence on minimum temperature. NINO 3.4 SST also shows moderate to strong positive correlations. Overall, the pattern suggests that during strong La Nina, PNA, IOD, and TNI play key roles in shaping minimum temperature variations.

Figure 25 shows how average temperature in different parts of Bangladesh relates to various climate factors during strong El Nino events. The x-axis outlines meteorological stations across Bangladesh, reflecting spatial changes in average temperature behavior, while the y-axis features the important climate indices—ONI, NINO 3.4 SST, SOI, IOD, PDO, BEST, PNA, TNI, and 200 mb Zonal Winds—that drive variations in T_{avg} across regions. The PDO has the strongest and most consistent positive relationship with temperature across all stations. NINO 3.4 SST, TNI, and upper-level winds also show clear positive links, especially in eastern regions. In contrast, the PNA index is linked with lower temperatures. Some indices like ONI, SOI, and BEST have little or no clear effect. The color scale makes it easy to compare the strength and direction of these relationships.

Figure 26 shows how maximum temperature (T_{max}) in Bangladesh correlates with climate indices during strong El Nino events. The x-axis represents different meteorological stations throughout Bangladesh, highlighting regional differences in maximum temperature responses. The y-axis lists major climate indices, including ONI, NINO 3.4 SST, SOI, IOD, PDO, BEST, PNA, TNI, and 200 mb Zonal Winds, which play a role in shaping T_{max} variations. NINO 3.4 SST, PDO, and TNI have positive moderate correlations across all stations.

*** In this section, four representative correlation heatmaps are presented, derived from four climate datasets—maximum temperature (T_{max}), minimum temperature (T_{min}), average temperature (T_{avg}), and rainfall, during four major ENSO events. These heatmaps illustrate the key relationships between ENSO indices and climate variables across Bangladesh. The remaining correlation heatmaps are provided in the annexure for completeness. Altogether, a total of 32 correlation heatmaps were generated, covering all combinations of the four climate variables and the identified ENSO events, enabling a comprehensive assessment of ENSO–climate interactions over the study area.***

Fig. 26 Correlation of ENSO Indices with $T_{\max}$ Across 29 Stations in Bangladesh during Strong El Nino Event



3.3 Feature engineering

To prepare the dataset for machine learning modeling, several preprocessing steps were implemented to ensure that the input features were both interpretable and suitable for numerical algorithms. These steps were necessary because climate datasets often contain categorical descriptors, varying measurement scales, and redundant variables, all of which can negatively affect model training if not addressed.

3.3.1 Categorical feature encoding

The dataset included categorical variables such as SEASON and ENSO Event, which required transformation into numerical format before being used in machine learning algorithms. One-hot encoding was applied to these variables using Scikit-learn's **OneHotEncoder**. Dropping the first category (drop = 'first') was essential to prevent multicollinearity, which occurs when redundant dummy variables provide overlapping information. This step was particularly important because tree-based models (RF, XGB) can overemphasize duplicated categorical information, and linear models (LR) can become unstable when perfect collinearity exists. Thus, encoding not only enabled numerical processing but also improved model interpretability and robustness.

3.4 Feature selection

After encoding, non-informative or operational columns such as Index, Phase Change, and Phase Duration were removed. These variables did not contribute meaningful predictive information and retained metadata rather than physical climate interpretation. Their inclusion would introduce noise and increase model complexity without improving accuracy. Separating the dataset into input features (x) and target variables (y) further ensured that only relevant predictors were used in model training. This step helped reduce overfitting and enhanced the interpretability of feature-importance measures used later in the analysis.

3.5 Data scaling

To standardize the numerical features and improve the performance of machine learning models, feature scaling was applied. Standard scaling was performed using the **StandardScaler** function from Scikit-learn, which transforms the data to have a mean of 0 and a standard deviation of 1. The scaler was fitted on the training data, and the same transformation was applied to the test data to maintain consistency. This step was essential because the dataset contains climate variables measured in different units and on varying scales—such as temperature ($^{\circ}\text{C}$), rainfall (mm), and multiple unitless climate indices—which could otherwise distort the learning process. Scaling prevents predictors with larger numeric ranges from disproportionately influencing model behavior, improves the convergence and stability of optimization-based algorithms such as XGBoost and Linear Regression, and ensures fair distance calculations in models like KNN. Additionally, standardized inputs help maintain numerical stability and prevent biased feature importance estimates. The scaler was fitted only on the training data to avoid data leakage, and the same transformation was applied to the test data, ensuring consistent and unbiased model evaluation.

Table 2 Performance Comparison of Machine Learning Models for Predicting Climate Variable (Training Dataset)

Performance Metrics	Dataset	Model Name					
		Random Forest Regressor	Decision Tree Regressor	K fold Cross validation	XGBoosting	Linear Regression	KNN Regressor
R2 Score	T _{max} (°C)	0.9700	0.7969	0.7751 ± 0.0318	0.9353	0.6525	0.7686
	T _{min} (°C)	0.9972	0.9788	0.9787 ± 0.0054	0.9956	0.9805	0.8998
	T _{avg} (°C)	0.9928	0.9526	0.9456 ± 0.0115	0.9846	0.8828	0.8624
	Rainfall (mm)	0.9464	0.6923	0.6216 ± 0.0235	0.8970	0.6302	0.6394
MAE	T _{max} (°C)	0.3443	0.9061	0.9423 ± 0.0599	0.5279	1.1833	0.9592
	T _{min} (°C)	0.1866	0.5313	0.5149 ± 0.0542	0.2455	0.5020	1.0857
	T _{avg} (°C)	0.2373	0.6118	0.6449 ± 0.0537	0.3571	1.0514	1.0583
	Rainfall (mm)	31.9164	75.7660	85.6627 ± 5.4976	47.8426	92.3468	89.6615
RMSE	T _{max} (°C)	0.4698	1.2256	1.2812 ± 0.1035	0.6860	1.6080	1.3114
	T _{min} (°C)	0.2556	0.7002	0.6917 ± 0.0785	0.3201	0.6745	1.5645
	T _{avg} (°C)	0.3253	0.8299	0.8764 ± 0.0720	0.4751	1.3160	1.4318
	Rainfall (mm)	51.6270	124.1331	135.9836 ± 7.6475	71.6564	137.1181	138.1089

Table 3 Performance Comparison of Machine Learning Models for Predicting Climate Variable (Testing Dataset)

Performance Metrics	Dataset	Model Name					
		Random Forest Regressor	Decision Tree Regressor	K fold Cross validation	XGBoosting	Linear Regression	KNN Regressor
R2 Score	T _{max} (°C)	0.8512	0.7918	0.7751 ± 0.0318	0.8509	0.6425	0.6095
	T _{min} (°C)	0.9693	0.9584	0.9787 ± 0.0054	0.9702	0.9679	0.8589
	T _{avg} (°C)	0.9521	0.9386	0.9456 ± 0.0115	0.9559	0.8472	0.7873
	Rainfall (mm)	0.6253	0.6146	0.6216 ± 0.0235	0.5897	0.5662	0.4325
MAE	T _{max} (°C)	0.7921	0.9195	0.9423 ± 0.0599	0.7889	1.1940	1.2683
	T _{min} (°C)	0.5144	0.6243	0.5149 ± 0.0542	0.5120	0.5529	1.2357
	T _{avg} (°C)	0.5719	0.6563	0.6449 ± 0.0537	0.5479	1.0951	1.2344
	Rainfall (mm)	94.3346	94.9789	85.6627 ± 5.4976	101.2635	107.4055	125.6424
RMSE	T _{max} (°C)	1.0180	1.2043	1.2812 ± 0.1035	1.0252	1.5982	1.6749
	T _{min} (°C)	0.7772	0.9042	0.6917 ± 0.0785	0.7678	0.7953	1.6955
	T _{avg} (°C)	0.7644	0.8630	0.8764 ± 0.0720	0.7316	1.3677	1.6149
	Rainfall (mm)	149.2662	152.6571	135.9836 ± 7.6475	154.7295	161.8352	188.6645

3.6 Model performance evaluation

Table 2 presents the performance comparison of six machine learning models (Random Forest, Decision Tree, K-Fold Cross-Validated model, XGBoost, Linear Regression, and KNN) for predicting different climate variables (T_{max}, T_{min}, T_{avg}, and Rainfall) using the training dataset. Overall, the Random Forest and XGBoost models consistently demonstrate higher R² values and lower error metrics, indicating superior predictive performance. The K-Fold cross-validated model provides stable and reliable results with low variance. In contrast, Linear Regression and KNN generally show weaker performance, especially for rainfall prediction, with higher MAE and RMSE values.

Table 3 presents the performance of six machine learning models (Random Forest Regressor, Decision Tree Regressor, K-Fold Cross Validation, XGBoosting, Linear Regression, and KNN Regressor) evaluated across four testing datasets (T_{max}, T_{min}, T_{avg}, and Rainfall). Three metrics are used: R² Score, MAE (Mean Absolute Error), and RMSE (Root Mean Square Error). The best-performing model is Random Forest in predicting T_{max}, with an R² score of 0.8512, the MAE of 0.7921, and RMSE of 1.0180. Similarly, Random Forest also excels in predicting Rainfall, achieving the highest R² score (0.6253), the MAE (94.3346), and the RMSE (149.2662). On the other hand, XGBoosting leads for T_{avg} dataset, with

an R^2 score of 0.9559, the MAE (0.5479), and RMSE of 0.7316. XGBoosting shows superior performance in $T_{\min}$ prediction, achieving the best balance across metrics with an R^2 score of 0.9702, an MAE of 0.5120, and RMSE of 0.7316.

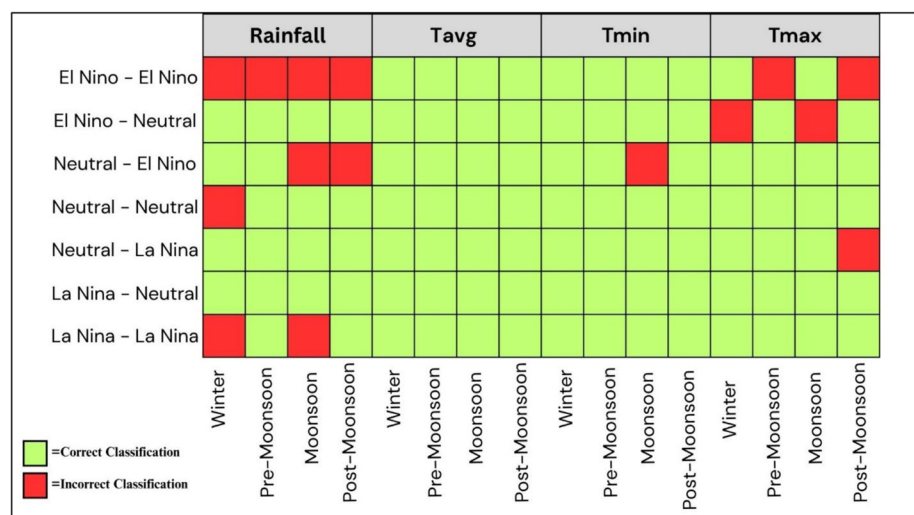
In the training phase (Table 2), Random Forest and XGBoost consistently achieve the highest R^2 scores and lowest error values, indicating strong learning capability and robustness, while the K-Fold cross-validated model ensures stable and reliable performance. In contrast, Linear Regression and KNN perform relatively poorly, particularly for rainfall prediction. The testing results in Table 3 confirm these trends. Random Forest shows superior performance for $T_{\max}$ and Rainfall, whereas XGBoost outperforms other models for $T_{\min}$ and T_{avg} , achieving the best balance between accuracy and error. The close agreement between training and testing performances indicates good generalization ability of the ensemble models, highlighting Random Forest and XGBoost as the most reliable and accurate approaches for climate variable prediction.

In particular, the XGBoosting model performed best for $T_{\min}$ and T_{avg} due to the consistency in these datasets. These datasets exhibit stable patterns, allowing XGBoost's gradient boosting approach to effectively capture and predict the underlying relationships with high precision. Its ability to handle non-linear relationships and outliers and fine-tune its performance through regularization further contributes to its success. The Random Forest Regressor excelled in predicting $T_{\max}$ and Rainfall, despite the dataset's random variability. Random Forest's ensemble approach, which combines predictions from multiple decision trees, effectively handles random variations and avoids overfitting. This robustness makes it ideal for datasets with less predictable patterns.

The Fig. 27 serves as a complementary visualization to Table 1 by translating the regression outputs into a classification-based interpretation. Each observation was categorized into three conceptual classes based on its deviation from the 10-year seasonal average: below-average, where actual or predicted values of rainfall, T_{avg} , $T_{\min}$, and $T_{\max}$ are lower than the corresponding average; above-average, where values exceed the 10-year seasonal mean; and neutral, where values lie close to the average and the probabilities of being above or below are nearly equal. The resulting heatmap employs a binary evaluation scheme in which green indicates correct classification, meaning the predicted class matches the observed class, while red denotes incorrect classification where the predicted and actual classes differ.

Critically, the model demonstrates strong alignment with actual trends, particularly for temperature variables (T_{avg} , $T_{\min}$, $T_{\max}$), where the predicted direction often matches the observed change. However, prediction accuracy appears more varied for rainfall, which shows several mismatches or uncertain classifications ("Both"), reflecting the complex and nonlinear nature of precipitation dynamics. Some combinations (e.g., La Nina–El Nino in Pre-Monsoon) show high concordance across all variables, while others (e.g., Neutral–Neutral in Post-Monsoon) reflect greater ambiguity.

Fig. 27 Comparison between Actual and Predicted Scenario under Different Conditions



4 Discussion

The results of this study provide a detailed understanding of how ENSO-related indices modulate Bangladesh's seasonal climate, but their interpretation must be situated within the context of earlier ENSO–rainfall studies in the region. Previous research has consistently shown that El Niño events are associated with suppressed monsoon rainfall and warmer winter temperatures, while La Niña events typically enhance monsoon rainfall and contribute to cooler winter conditions in Bangladesh [11, 35]. The correlations found in this study, especially reduced rainfall during El Niño phases and increased rainfall during La Niña, are highly consistent with these well-established teleconnection patterns.

Where this study advances earlier work is in demonstrating that several non-traditional indices (e.g., PDO, TNI, PNA) show correlations equal to or stronger than the canonical Niño 3.4 or ONI. This aligns with emerging evidence that ENSO's regional impacts are increasingly shaped by extratropical and Indian Ocean processes rather than solely by central Pacific SST anomalies [9, 13]. For instance, the consistently strong influence of PDO on $T_{\max}$, $T_{\min}$, and T_{avg} observed in this study supports the understanding that Pacific decadal variability can modulate the strength and spatial pattern of ENSO teleconnections. Similar behavior has been reported in South Asian monsoon studies, which show that warm PDO phases amplify El Niño drying, while cool PDO phases can weaken it [3, 21].

The results also reveal that IOD and TNI frequently exhibit strong negative correlations with rainfall and temperature during La Niña phases. This is physically consistent with their documented roles in modulating moisture transport and convective activity across the Indo-Pacific region [5, 9]. A positive IOD event, for example, may counteract El Niño–induced drying by enhancing westerly moisture flow into the Bay of Bengal, while negative IOD conditions often reinforce La Niña–related cooling and precipitation anomalies across South Asia [3].

Furthermore, the strong positive correlation between PNA and temperature variables ($T_{\max}$, $T_{\min}$, T_{avg}) observed in multiple ENSO phases suggests an important mid-latitude influence on Bangladesh's climate. PNA's documented role in generating Rossby wave trains that modulate upper-level zonal winds and wintertime temperature variability [13] supports this finding. The significant involvement of 200 mb zonal winds in the model also reinforces the idea that upper-tropospheric circulation anomalies contribute meaningfully to seasonal variability over South Asia.

Another important contribution of this study is the spatial differentiation of teleconnections across Bangladesh. Stations in the northeast and southeast exhibited stronger correlations with Pacific indices, consistent with prior findings that monsoon trough positioning and Bay of Bengal moisture influx are more sensitive in these regions [11]. In contrast, weaker teleconnections in the northwest align with earlier climatological studies showing greater continental influence and reduced monsoon penetration [1]. These results confirm that ENSO teleconnections are both seasonally and geographically modulated within Bangladesh.

When compared with prior work, the exceptionally high model skill for temperature ($T_{\min}$ $R^2 = 0.9693$, T_{avg} $R^2 = 0.9559$, $T_{\max}$ $R^2 = 0.8512$) is consistent with the broader literature: temperature anomalies tend to be more directly controlled by large-scale radiative and circulation drivers and are therefore easier for multi-index, nonlinear models to predict than precipitation [24]. When benchmarked against prior work, the rainfall R^2 obtained in this study (0.6253) aligns well with expectations for a spatially extensive, multi-decadal national dataset. Earlier ENSO–rainfall studies in Bangladesh typically report modest explanatory power when using single indices, often corresponding to R^2 values below 0.30 [11]. In contrast, localized or basin-specific machine-learning studies frequently achieve higher predictive skill, with Random Forest, Gradient Boosting, or LSTM models reporting R^2 values in the range of 0.70–0.80 for monthly or seasonal rainfall [15, 39]. These elevated values, however, are generally derived from smaller spatial domains, shorter temporal windows, and more homogeneous hydroclimatic regimes, which inherently reduce variability and increase model fit. Therefore, the rainfall R^2 obtained in this study represents a robust and realistic estimate for a heterogeneous 29-station dataset spanning 45 years. Moreover, the comparatively higher skill observed for temperature variables ($T_{\min}$, T_{avg} , and $T_{\max}$) is consistent with established climatological understanding: temperature responds more directly to large-scale radiative forcing and circulation anomalies, whereas rainfall is influenced by additional mesoscale and intraseasonal processes that limit predictive skill even in advanced ML frameworks.

Overall, the findings show that ENSO teleconnections affecting Bangladesh arise from a combination of tropical Pacific, Indian Ocean, and extratropical circulation modes. Unlike earlier Bangladesh-focused studies that relied primarily on single indices such as Niño 3.4 or SOI, this multi-index approach captures a more comprehensive set of physical drivers, including decadal variability and cross-basin interactions. The strong performance of ensemble ML models, Random Forest and XGBoost, further demonstrates the advantage of nonlinear, multi-predictor frameworks

for diagnosing complex climate relationships [28]. By integrating nine teleconnection indices and applying interpretable machine learning, this study offers a more holistic explanation for why ENSO impacts vary across seasons, ENSO phases, and regions within Bangladesh.

5 Conclusion

This study explored the influence of ENSO-related climate indices on temperature and rainfall patterns across Bangladesh using 45 years of observational data and a suite of supervised machine learning models. Strong associations between regional weather parameters and indices such as NINO 3.4 SST, SOI, PDO, and TNI were identified, particularly during pronounced El Niño and La Niña phases. Among the evaluated models, XGBoost delivered the highest predictive accuracy for $T_{\min}$ ($R^2 = 0.9702$) and T_{avg} ($R^2 = 0.9559$), while Random Forest performed best for $T_{\max}$ ($R^2 = 0.8512$) and rainfall ($R^2 = 0.6253$), demonstrating the capability of ensemble methods to capture complex climate relationships.

Seasonal forecasts generated using the best-performing models revealed consistent warming trends in $T_{\min}$ and T_{avg} across most ENSO transitions, with notable declines in $T_{\max}$ during monsoon periods. Rainfall predictions were more variable, showing increases during La Niña-related transitions and declines during El Niño-associated phases, reflecting the inherent complexity of precipitation dynamics. These findings underscore the practical value of machine learning-based forecasting for agricultural planning, water resource management, and disaster preparedness, enabling more informed decision-making in climate-sensitive sectors.

Future work should prioritize the integration of high-resolution remote sensing datasets, the inclusion of a broader set of teleconnection indices, and the implementation of advanced deep learning architectures—such as LSTM networks and hybrid ensemble frameworks, to further enhance predictive performance. The relatively lower R^2 values obtained for rainfall prediction can be attributed to the study's exclusive reliance on ENSO indices, even though rainfall variability is also strongly influenced by other large-scale atmospheric phenomena, such as the Madden–Julian Oscillation. We acknowledge this limitation and recommend that future research incorporate additional atmospheric drivers to capture rainfall dynamics more comprehensively. Furthermore, extending the modeling framework to simulate extreme weather events, including heatwaves, droughts, and floods, would strengthen early warning systems and support long-term climate resilience planning for Bangladesh and other ENSO-sensitive regions.

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Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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